PolymerFFKit

Polymer Force Field Parameter Fitting and Topology Building Tool Software

(Standard Screenshot Manual)

Version: V1.0.0

Date: May 2026

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1. Introduction

1.1 Purpose

This document is the Standard Screenshot Manual for PolymerFFKit (Polymer Force Field Parameter Fitting and Topology Building Tool Software). It is prepared to support the software copyright registration application by visually demonstrating the functional interfaces and operational workflows of the software through screenshots. This manual is intended for software copyright reviewers to understand the functional composition and user interaction methods of the software.

1.2 Background

PolymerFFKit is a force field parameter fitting and topology building tool designed for polymer molecular dynamics simulations. Independently developed, it provides molecular dynamics researchers with a complete workflow from all-atom trajectories to coarse-grained force field parameters, encompassing trajectory mapping, geometric analysis, Boltzmann inversion fitting, non-bonded parameter assignment, polymer sequence generation, and three-dimensional structure and topology building.

The software adopts a command-line interactive interface (Terminal User Interface, TUI) built on the Python Rich library, providing user-friendly terminal menu navigation and interactive input prompts with Tab path auto-completion support.

2. Software Overview

2.1 Features

PolymerFFKit consists of two major functional modules:

(1) Trajectory Analysis and Force Field Generation Module: This module includes four steps: molecular mapping (Step 11, supporting AA/UA/CG resolutions), trajectory geometry extraction and analysis (Step 12, automatic identification of bond length, angle, and dihedral distributions), Boltzmann inversion and force field fitting (Step 13, converting frequency distributions to potential energy curves with harmonic/Fourier/RB fitting), and non-bonded parameter assignment (Step 14, interactive assignment of Lennard-Jones parameters and partial charges).

(2) Polymer Sequence and Topology Builder Module: This module includes two steps: polymer sequence generation (Step 21, supporting precise-ratio random copolymer sequence generation) and 3D structure and topology building (Step 22, constructing 3D atomic coordinates using the NERF algorithm and outputting files in PDB, GROMACS ITP/TOP, and LAMMPS DATA formats).

The main menu interface of the software is shown in Figure 1.

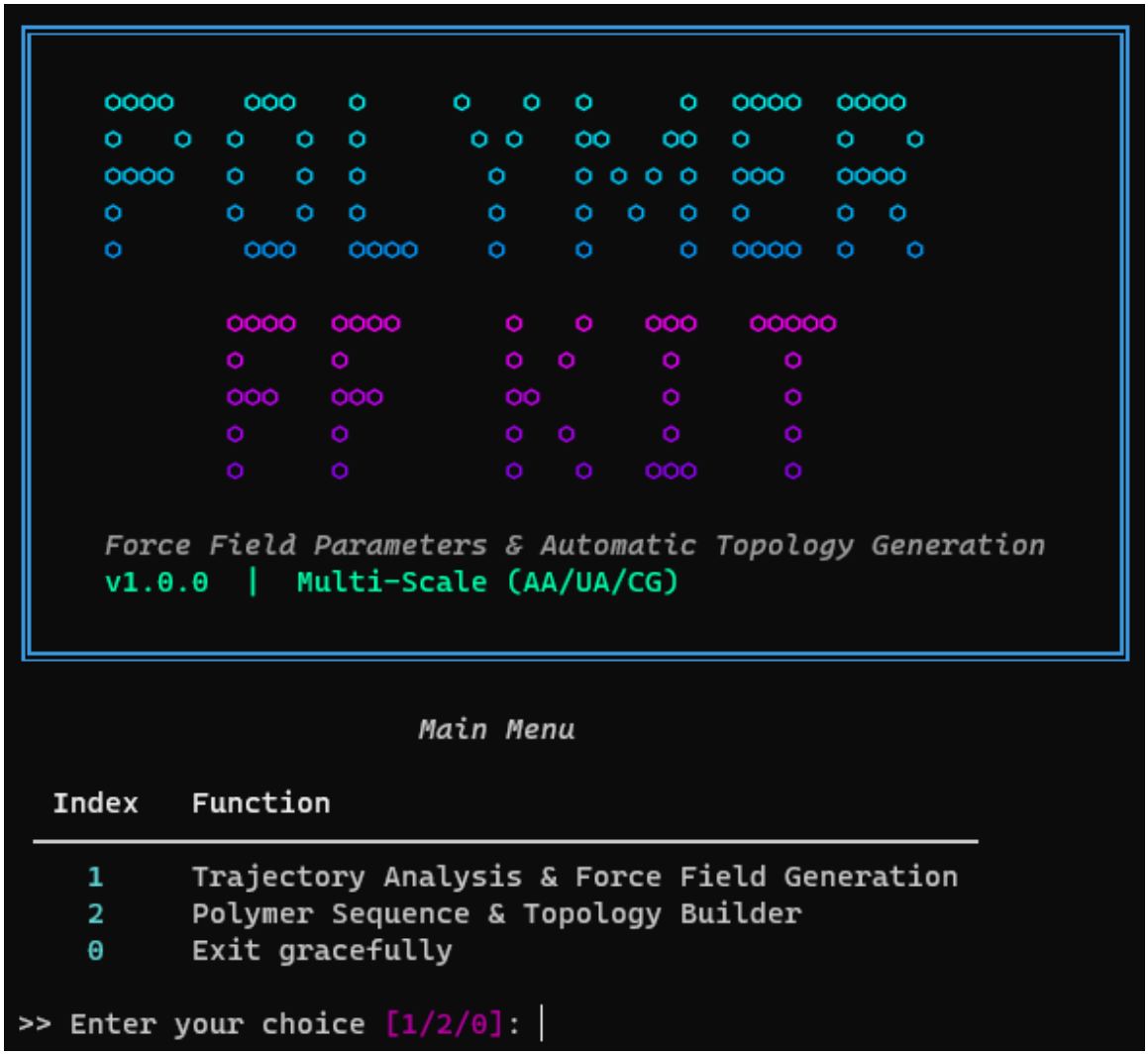


Figure 1 Program Startup -- Main Menu Interface

2.2 Performance

The software delivers stable performance, utilizing vectorized numerical computation (NumPy) and scientific fitting algorithms (SciPy) to efficiently process large-scale molecular dynamics trajectory data. It supports output in multiple force field formats, meeting the requirements of different molecular dynamics simulation software (GROMACS, LAMMPS).

3. Operating Environment

3.1 Hardware Requirements

CPU: Intel / AMD x86-64 architecture processor

Memory: 16 GB RAM or above

Storage: 512 GB SSD or above

Display: 1920x1080 or higher resolution

3.2 Software Requirements

Operating System: Windows 10/11, Ubuntu 22.04+, macOS 13+

Runtime Environment: Python 3.10 or above

Dependencies: NumPy, SciPy, Pandas, Matplotlib, Rich, prompt_toolkit, etc.

External Software: GROMACS (optional, for force field validation), LAMMPS (optional, for force field validation)

4. Software Interface Screenshots and Descriptions

This software uses a command-line interactive interface (Terminal User Interface, TUI). All operations are performed within the terminal. The following sections present screenshots and descriptions of each functional interface in order of the modules.

4.1 Program Startup and Main Menu Interface

Upon program startup, the terminal displays the ASCII-art-style launch banner of PolymerFFKit, featuring a cyan-gradient 'POLYMER FF KIT' logo, followed by the version number (V1.0.0) and the multi-scale identifier (AA/UA/CG). The main menu offers three options:

(1) Trajectory Analysis & Force Field Generation

(2) Polymer Sequence & Topology Builder

(0) Exit gracefully

Users select the desired function by entering 1, 2, or 0. The main menu interface is shown in Figure 1. Additionally, users may press Ctrl+C at any time to safely interrupt the program.

4.2 Force Field Parameter Derivation Module

After selecting '1' in the main menu, the program enters the Force Field Parameter Derivation sub-module. This is one of the core functionalities of PolymerFFKit, providing a complete workflow from all-atom trajectories to force field parameters.

4.2.1 Force Field Parameter Derivation Submenu

This submenu displays four step options:

Step 11: United-atom or Coarse-Graining Mapping (AA to UA/CG)

Step 12: Trajectory Extraction and Analysis (Trajectory to CSV)

Step 13: Boltzmann Inversion & Fit (CSV to Bonded Constants)

Step 14: Assign Non-Bonded Params (Topology-driven Assign)

-1: Return to Main Menu

The Force Field Parameter Derivation submenu interface is shown in Figure 2.

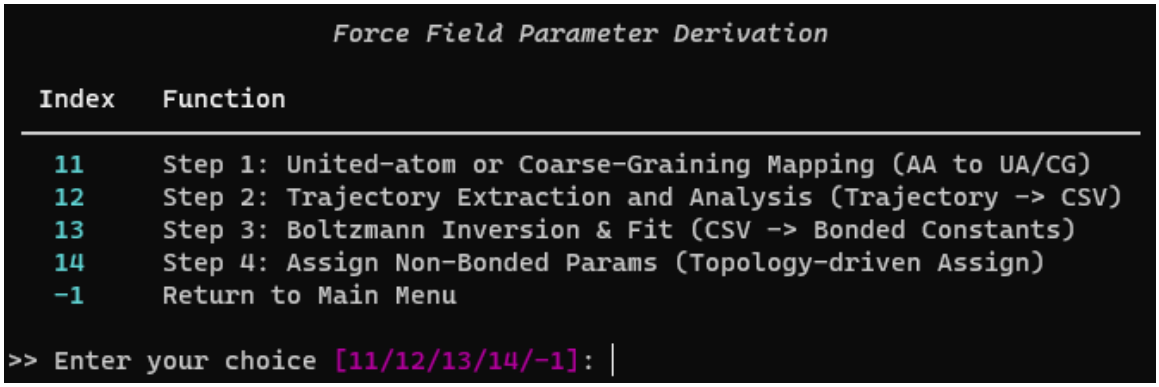


Figure 2 Force Field Parameter Derivation Submenu Interface

4.2.2 Step 11: Trajectory Mapping and Extraction

This step implements molecular trajectory resolution conversion, one of the core algorithms of the software. The operational workflow is as follows:

(1) Select target mapping resolution (AA/UA/CG)

(2) Input the all-atom trajectory file path (.xyz format, with Tab auto-completion)

(3) The system automatically scans and displays the total number of trajectory frames

(4) Enter the percentage of the trajectory to process (e.g., 80 for the last 80%)

(5) For UA mode: choose 'auto' (automatic hydrogen-to-heavy-atom collapse) or 'custom' mapping rules

(6) For CG mode: provide a mapping rules JSON file

(7) Select center calculation method (COM = center of mass, COG = center of geometry)

(8) Specify the output trajectory file path

The system then begins trajectory processing, outputting progress information every frame. The operational interface of Step 11 is shown in Figure 3.

```
>> Enter your choice [11/12/13/14/-1]: 11

============================================================
[Step 1] Trajectory Mapping & Extraction
============================================================
>> Select target mapping resolution [AA/UA/CG] (CG): CG
[*] Selected Mode: CG
(Tip: Press TAB to auto-complete paths, or drag & drop the file here)
step11_aa_trajectory.xyz

Scanning trajectory to count total frames...
[*] Total frames found: 3
>> Enter the percentage of trajectory to process (e.g., 80 for last 80%) (100): 100
[*] Will process 3 frames (starting from frame 0).

step11_cg_mapping_rules.json
[+] Loaded 2 mapping rules.
>> Select center logic (1: COM, 2: COG) [1/2] (1): 1
```

Figure 3 Step 11: Trajectory Mapping -- Operational Interface

Upon completion, the system displays a 'Task Complete' success message, including the number of processed frames, the output file path, and the total elapsed time. The completion interface of Step 11 is shown in Figure 4.

```
Starting trajectory processing...


[Success] Task Complete! 3 frames saved to ./CG_mapped.xyz
Time elapsed: 0.00 seconds.

                    Force Field Parameter Derivation

  Index   Function
 ──────────────────────────────────────────────────────────────────
   11     Step 1: United-atom or Coarse-Graining Mapping (AA to UA/CG)
   12     Step 2: Trajectory Extraction and Analysis (Trajectory -> CSV)
   13     Step 3: Boltzmann Inversion & Fit (CSV -> Bonded Constants)
   14     Step 4: Assign Non-Bonded Params (Topology-driven Assign)
   -1     Return to Main Menu

>> Enter your choice [11/12/13/14/-1]:
```

Figure 4 Step 11: Trajectory Mapping -- Completion Interface

4.2.3 Step 12: Trajectory Geometry Extraction and Analysis

This step automatically extracts bond length, bond angle, and dihedral angle geometric distribution data from the mapped trajectory file. The operational workflow is as follows:

(1) Input the mapped trajectory file path (.xyz format)

(2) Specify the output raw analysis data file path (default: ./raw_analysis.txt)

(3) Enter the bond cutoff distance (BOND_CUTOFF), with recommended values of 0.8 nm for CG and 0.2 nm for AA

The system then scans and analyzes all frames: it uses a distance-cutoff-based adjacency matrix to automatically identify chemical bond connectivity, and exhaustively computes all bond lengths, bond angles, and dihedral angles for each frame. Progress is reported every 100 frames. The operational interface of Step 12 is shown in Figure 5.

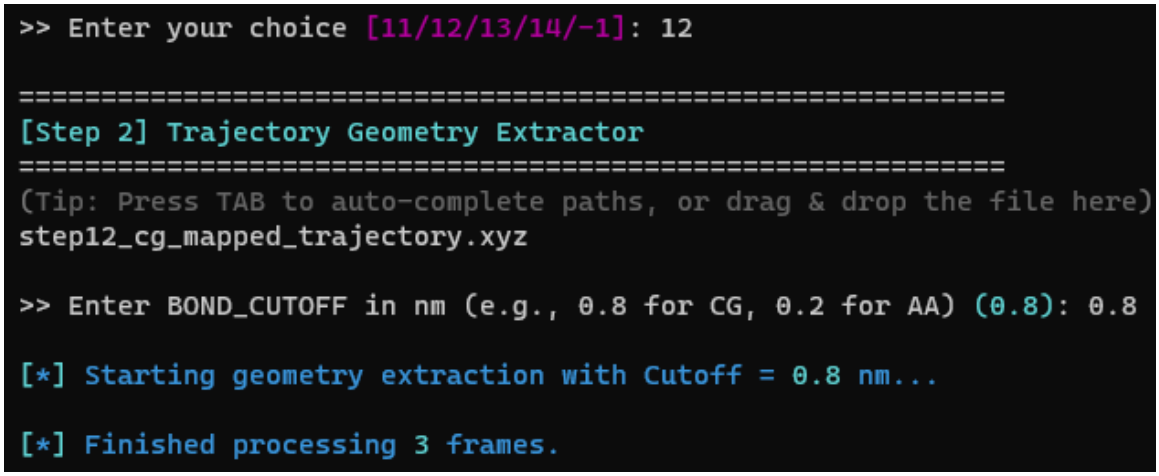


Figure 5 Step 12: Trajectory Geometry Extraction -- Operational Interface

After processing completes, the system displays statistics on identified geometry types (number of bond types, angle types, dihedral types) and outputs the raw analysis data file, which preserves all measured values for each bond/angle/dihedral type. The completion interface of Step 12 is shown in Figure 6.

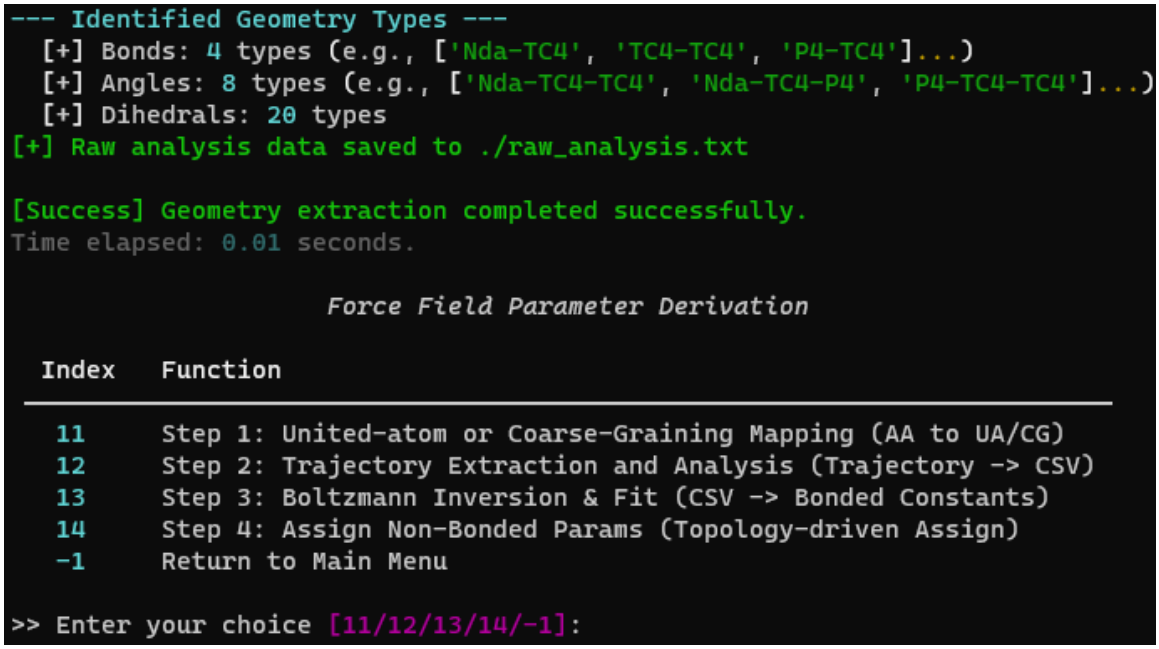


Figure 6 Step 12: Trajectory Geometry Extraction -- Completion Interface

4.2.4 Step 13: Boltzmann Inversion and Force Field Fitting

This step converts the geometric distribution data extracted in Step 12 into a Potential of Mean Force (PMF) via Boltzmann inversion and performs force field function fitting. The operational workflow is as follows:

(1) Input the raw analysis data file from Step 12 (CSV or text format)

(2) Specify the output directory (default: ./fit_results)

(3) Set bond fitting parameters: temperature (default 300 K) and PMF threshold percentage (default 50%)

(4) Set angle fitting parameters: temperature (default 300 K) and PMF threshold percentage (default 50%)

(5) Select dihedral fitting type: 0 = skip, 1 = periodic Fourier series (order 1-4), 2 = Ryckaert-Bellemans function

(6) Choose whether to generate PMF fitting plots (default: yes)

The operational interface of Step 13 is shown in Figure 7.

```
>> Enter your choice [11/12/13/14/-1]: 13

============================================================
[Step 3] Boltzmann Inversion & FF Fitter
============================================================
(Tip: Press TAB to auto-complete paths, or drag & drop the file here)
step13_analysis_data.csv


--- Bond Parameters ---
>> Temperature (K) (300.0): 300.0
>> PMF Threshold (% of Y-axis to fit) (50.0): 50.0

--- Angle Parameters ---
>> Temperature (K) (300.0): 300.0
>> PMF Threshold (% of Y-axis to fit) (50.0): 50.0

--- Dihedral Parameters ---
>> Fit dihedrals? [0/1/2] (0): 0

>> Generate PMF fit plots? (y/n) [y/n] (y): y

[*] Reading data and fitting Boltzmann Distributions...
  [+] Fitted bond_TC4-Nda successfully.
  [+] Fitted bond_Nda-TC4 successfully.
  [+] Fitted angle_TC4-Nda-TC4 successfully.
```

Figure 7 Step 13: Boltzmann Inversion and Force Field Fitting -- Operational Interface

The system sequentially performs PMF calculation, noise filtering, and curve fitting for each geometry type. Fitting results are saved as a text report file (forcefield_params.txt) containing equilibrium values (x_0) and force constants (k) for bonds and angles, as well as order-specific parameters for dihedrals. If plotting is enabled, each fitted item generates a PNG-format fitting curve plot. The completion interface of Step 13 is shown in Figure 8.

```
[Success] Fitting complete! Text report saved to: ./fit_results\forcefield_params.txt
[+] Fit plots saved to directory:
E:\from_computer\OneDrive\Desktop\important\PolymerFFKit-tools\软件著作权登记资料\input_templates\fit_results/

                    Force Field Parameter Derivation

  Index   Function
 ------------------------------------------------------------------
   11     Step 1: United-atom or Coarse-Graining Mapping (AA to UA/CG)
   12     Step 2: Trajectory Extraction and Analysis (Trajectory -> CSV)
   13     Step 3: Boltzmann Inversion & Fit (CSV -> Bonded Constants)
   14     Step 4: Assign Non-Bonded Params (Topology-driven Assign)
   -1     Return to Main Menu

>> Enter your choice [11/12/13/14/-1]:
```

Figure 8 Step 13: Boltzmann Inversion Fitting -- Completion Interface

4.2.5 Step 14: Non-Bonded Parameter Assignment

This step integrates the bonded parameters obtained from fitting with non-bonded parameters (Lennard-Jones parameters and partial charges) from the force field library into a unified force field configuration file. The operational workflow is as follows:

(1) Input the mapping rules JSON file path (from Step 11)

(2) Input the fitted parameters TXT file path (from Step 13)

(3) Specify the output unified configuration JSON file path

(4) Select the target resolution (AA/UA/CG)

(5) Select the reference L-J force field library: choose from built-in libraries (e.g., Martini 3.0), or provide a custom ITP file, or skip for manual input only

(6) Optional: load an external charge file (e.g., RESP/AM1-BCC computed partial charges)

The operational interface of Step 14 is shown in Figure 9.

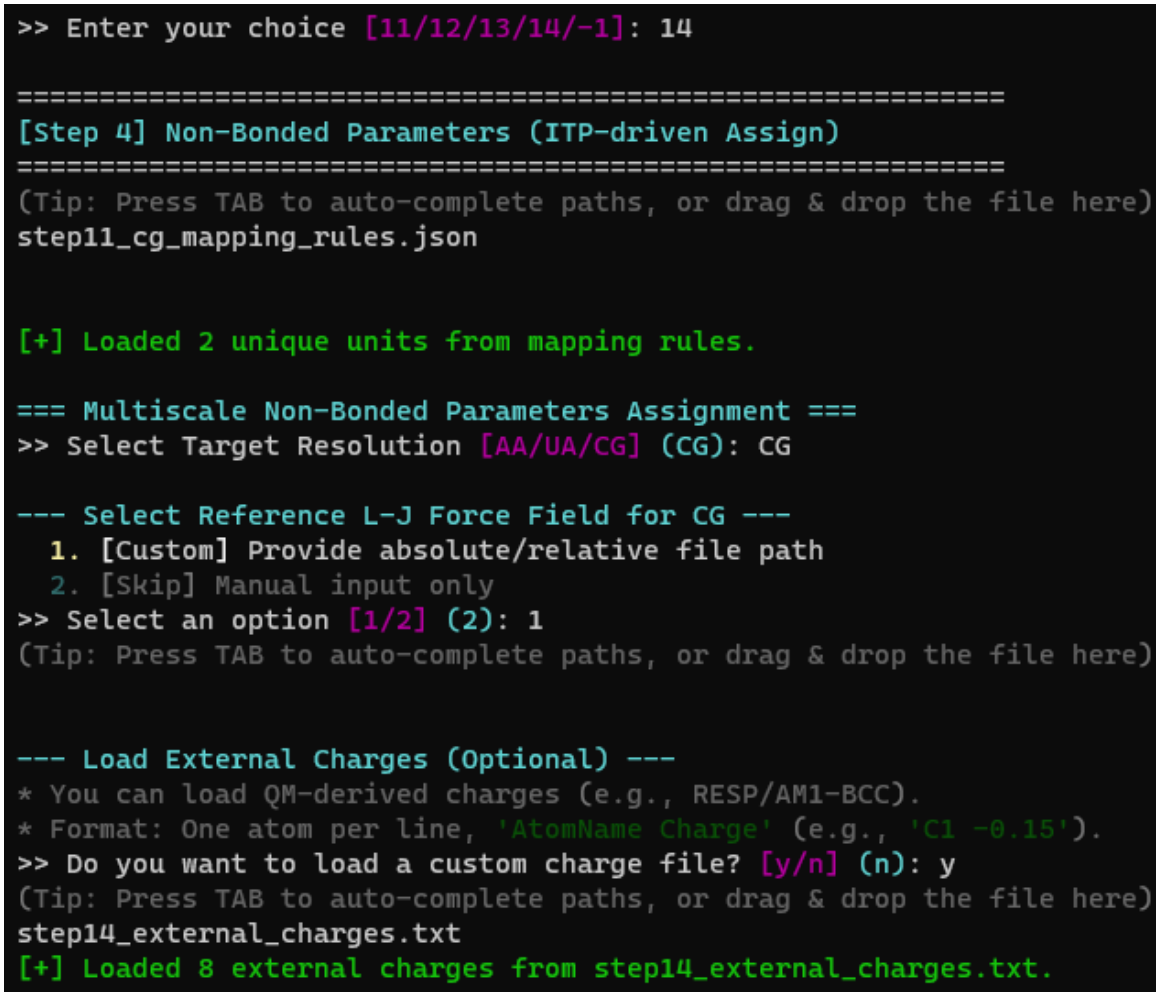


Figure 9 Step 14: Non-Bonded Parameter Assignment -- Force Field Library Selection Interface

The program then enters the interactive bead parameter assignment phase. The system displays each bead name to be assigned one by one. The user specifies the force field type, and the system automatically retrieves the corresponding L-J parameters (sigma, epsilon) and mass from the ITP database. If the type is not found in the database, manual input mode is entered. The user can confirm or edit the charge value and set the mass for each bead. The bead parameter assignment interface of Step 14 is shown in Figure 10.

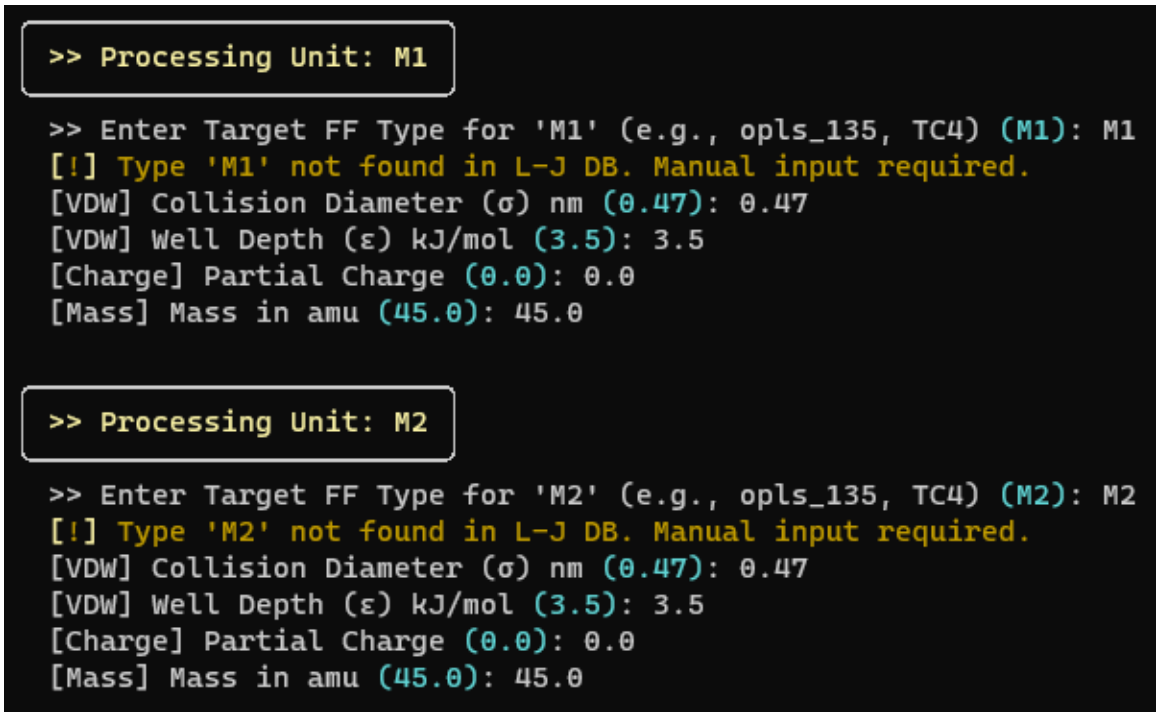


Figure 10 Step 14: Interactive Bead Parameter Assignment Interface

After all beads are assigned, the system displays a summary table of final assigned parameters (including bead name, force field type, mass, charge, and L-J source), and exports the complete configuration as a unified force field configuration JSON file (ff_unified_config.json) for subsequent topology building steps. The completion interface of Step 14 is shown in Figure 11.

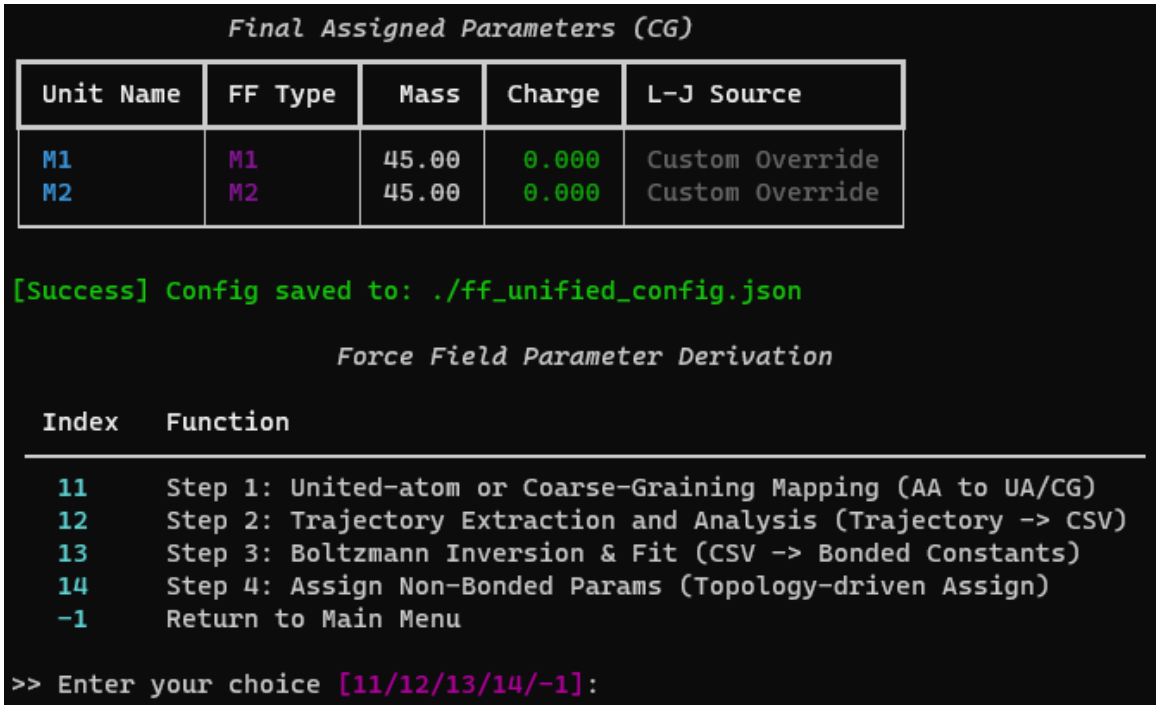


Figure 11 Step 14: Non-Bonded Parameter Assignment -- Completion Interface

4.3 Polymer Sequence and Topology Builder Module

After selecting '2' in the main menu, the program enters the Polymer Sequence and Topology Builder module. This module provides a complete workflow from sequence design to 3D structure construction.

4.3.1 Builder Submenu

This submenu displays two step options:

Step 21: Generate Polymer Sequence (Config & Randomization)

Step 22: Build 3D Structure & Topology (PDB/ITP/DATA)

-1: Return to Main Menu

The Builder submenu interface is shown in Figure 12.

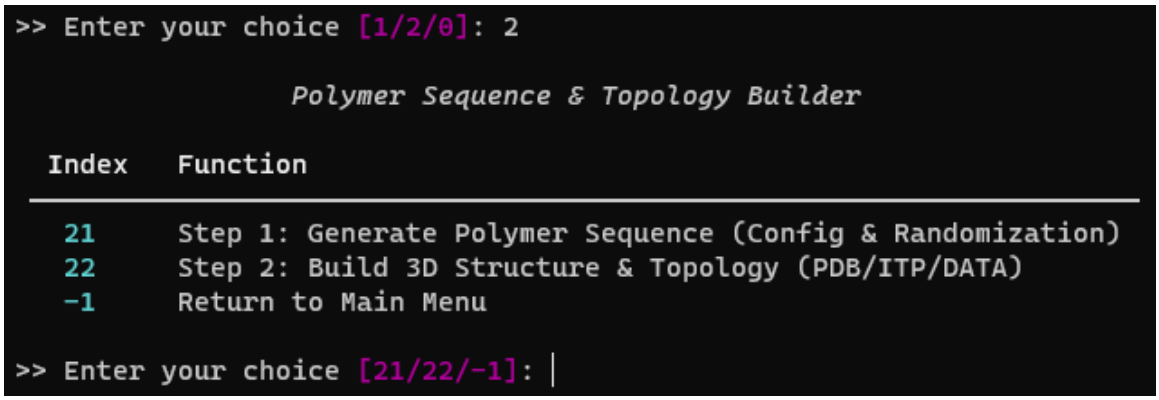


Figure 12 Polymer Sequence and Topology Builder Submenu Interface

4.3.2 Step 21: Polymer Sequence Generation

This step generates random copolymer sequences based on user-specified monomer types, ratios, and degree of polymerization. The operational workflow is as follows:

(1) Optionally load parameters from an existing JSON configuration file, or manually input

(2) In manual mode: enter repeat unit names (comma-separated, e.g., C, T, V)

(3) Enter the corresponding ratios for each unit (e.g., 5.5, 0.5, 4.0)

(4) Enter the total degree of polymerization (DP, default: 100)

(5) Enter the number of sequences to generate (default: 3)

(6) Choose whether to use strict mode to ensure exact monomer ratios

(7) Specify the output file prefix

The operational interface of Step 21 is shown in Figure 13.

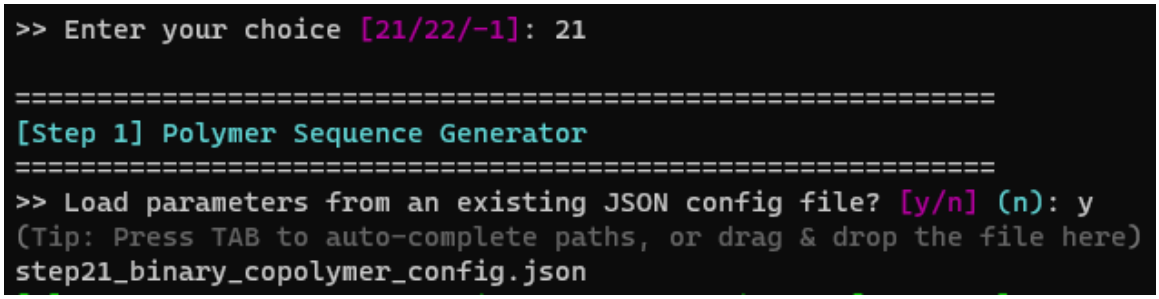


Figure 13 Step 21: Polymer Sequence Generation -- Operational Interface

Upon generation completion, the system outputs three files: (1) a FASTA-format sequence file (_sequences.txt); (2) a sequence composition analysis file (_analysis.txt, containing actual percentages of each monomer in each sequence); and (3) a JSON configuration file (_config.json, preserving parameters for reproducibility). The completion interface of Step 21 is shown in Figure 14.

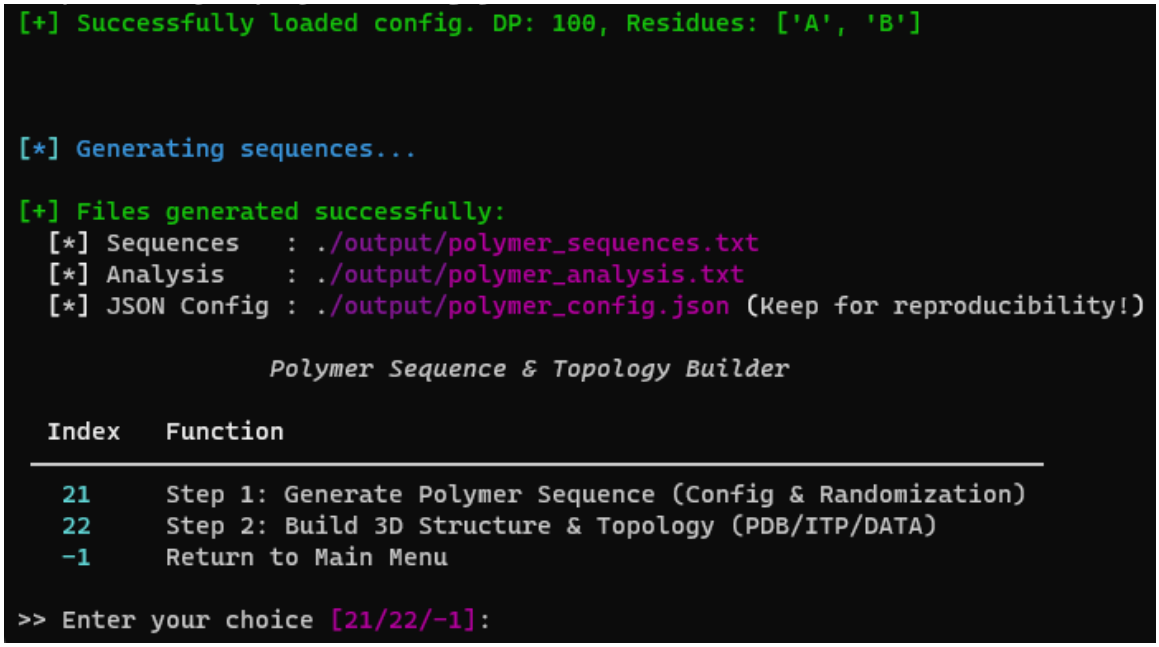


Figure 14 Step 21: Sequence Generation -- Completion Interface

4.3.3 Step 22: 3D Structure and Topology Building

This step is one of the core functionalities of PolymerFFKit. It converts one-dimensional polymer sequences into three-dimensional atomic coordinates and generates topology files for multiple molecular dynamics software packages. The operational workflow is as follows:

(1) Input the sequence file path generated in Step 21 (.txt format)

(2) Input the unified force field configuration JSON file path (from Step 14)

(3) Input the non-bonded parameter library ITP file path (e.g., Martini force field library)

(4) Specify the output directory (default: ./topology_out)

(5) Enter the number of molecules (for the GROMACS .top file)

The operational interface of Step 22 is shown in Figure 15.

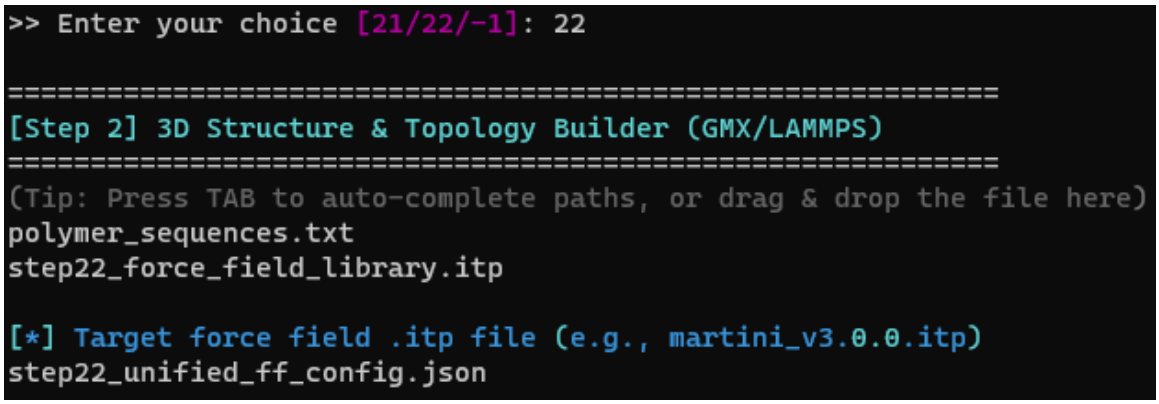


Figure 15 Step 22: 3D Structure and Topology Building -- Operational Interface

After loading the configuration, the system uses the NERF (Nearest Neighbor Effect Resolution Function) algorithm to construct complete 3D atomic coordinates from the 1D sequence: the first atom is placed at the origin, the second atom along the X-axis, and subsequent atoms are positioned step by step based on bond length, bond angle, and dihedral angle parameters. Main-chain atoms are arranged linearly, and branch atoms follow branch definitions and dihedral offset rules. For each sequence, the system generates: a PDB file (CHAIN_N.pdb, with CONECT bond records), a LAMMPS data file (CHAIN_N.data, containing Masses, Bond Coeffs, Angle Coeffs, Atoms, Bonds, and

Angles sections), and merged GROMACS ITP/TOP files. The processing and completion interface of Step 22 is shown in Figure 16.

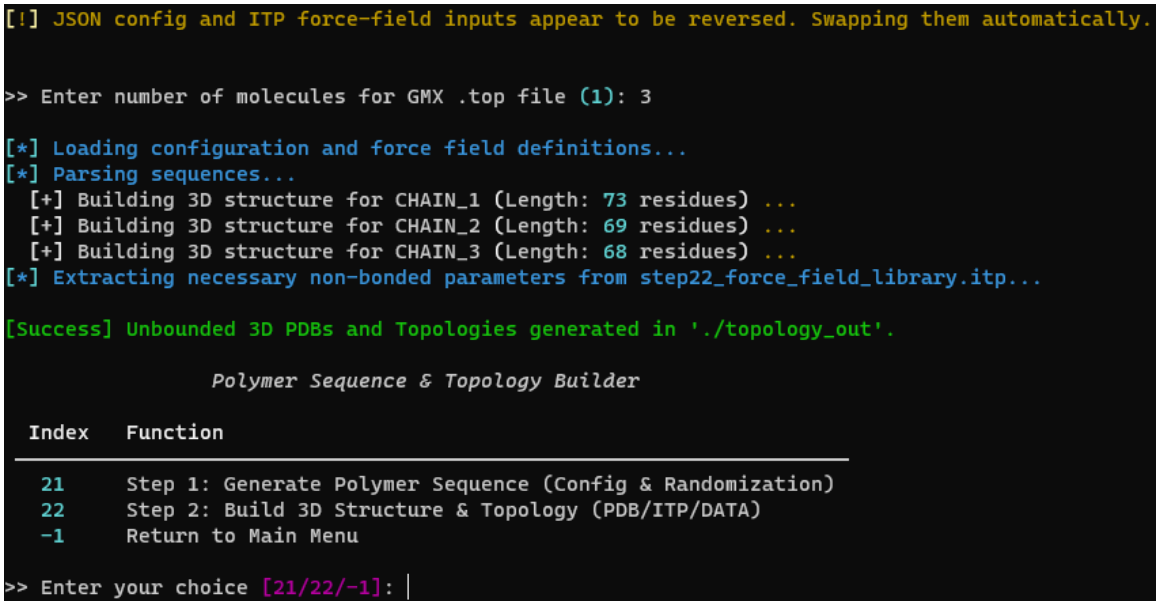


Figure 16 Step 22: 3D Structure and Topology Building -- Completion Interface

4.4 Program Exit Interface

After selecting '0' in the main menu, the program displays the exit message 'Exiting PolymerFFKit. Have a great day!' and terminates normally. Users may also press Ctrl+C at any time to interrupt the program; the program will display 'Program interrupted by user. Exiting...' before exiting safely. The program exit interface is shown in Figure 17.

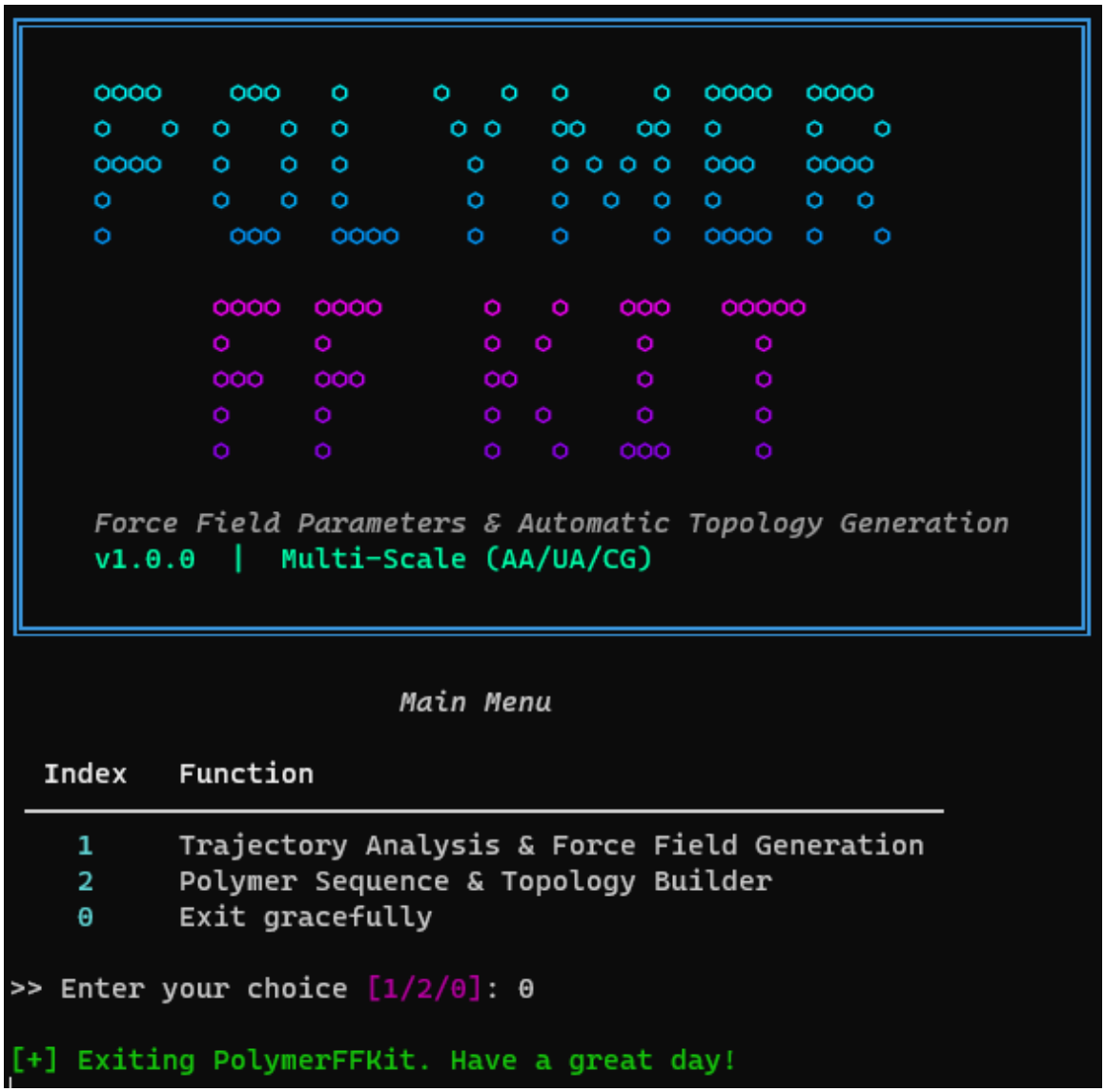


Figure 17 Program Normal Exit Interface